

E9 205 Machine Learning for Signal Processing

Neural Networks - Generalization

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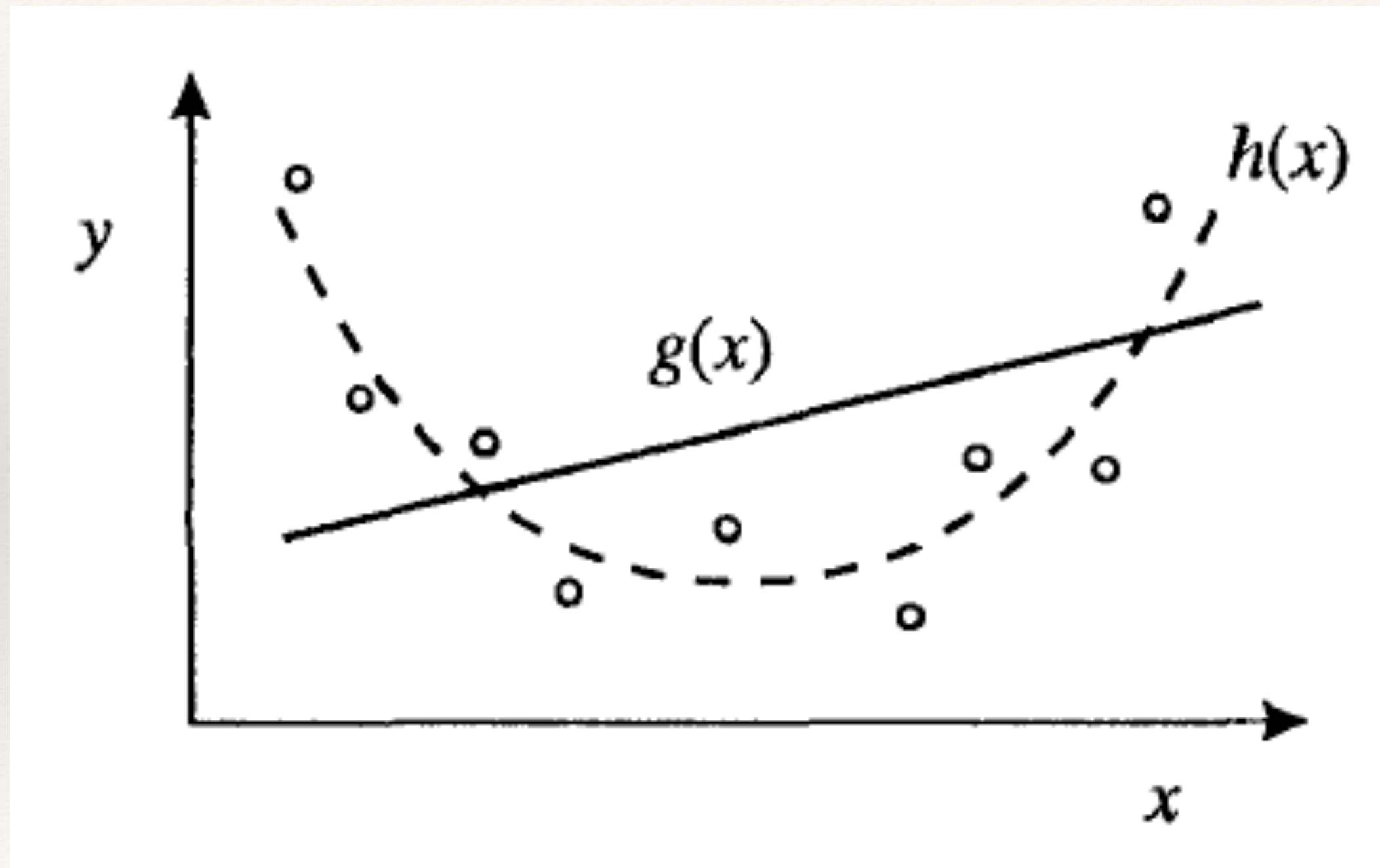


Bias and Variance In Neural Network Training

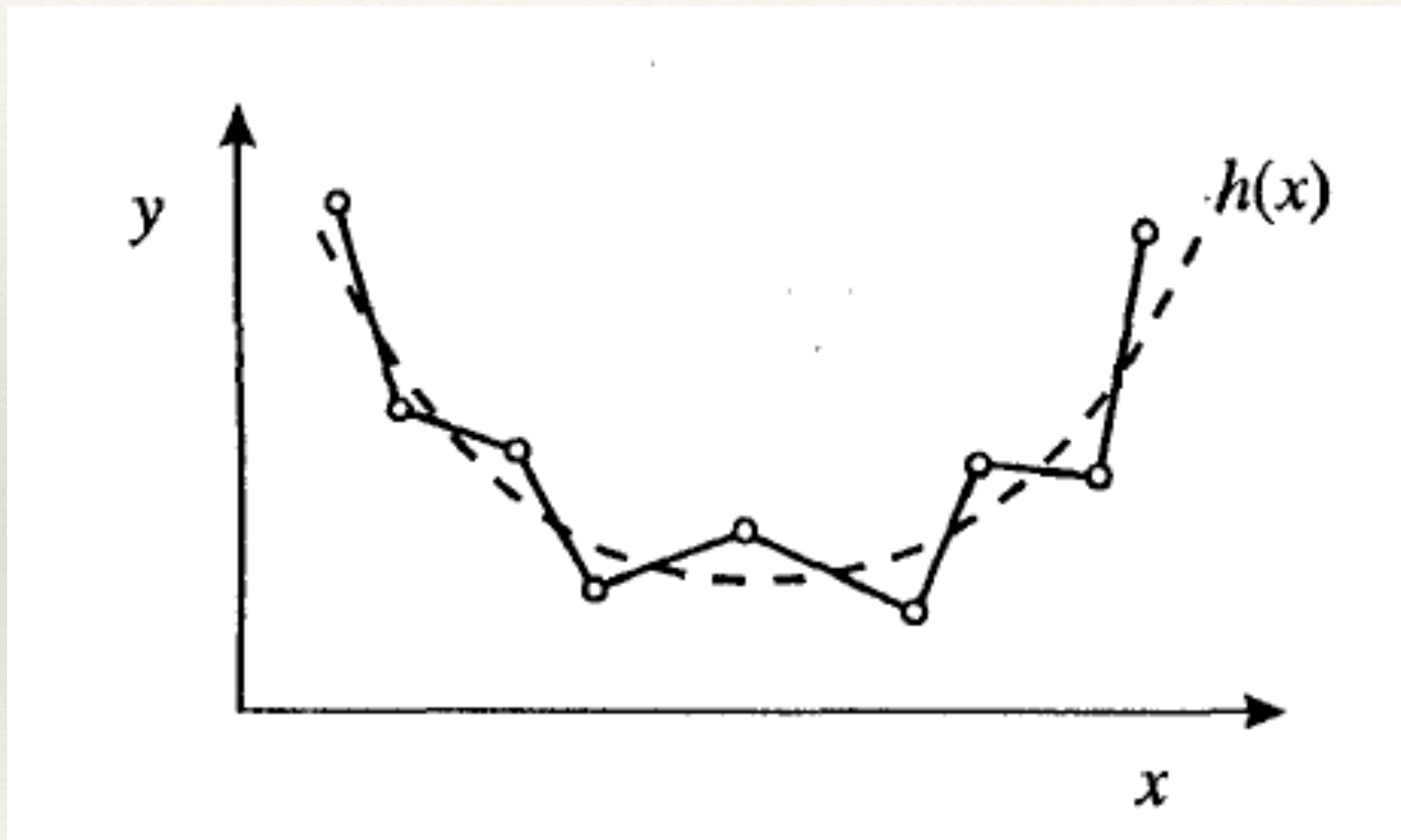
$$(\text{bias})^2 = \frac{1}{2} \int \{\mathcal{E}_D[y(\mathbf{x})] - \langle t|\mathbf{x} \rangle\}^2 p(\mathbf{x}) d\mathbf{x}$$

$$\text{variance} = \frac{1}{2} \int \mathcal{E}_D[\{y(\mathbf{x}) - \mathcal{E}_D[y(\mathbf{x})]\}^2] p(\mathbf{x}) d\mathbf{x}.$$

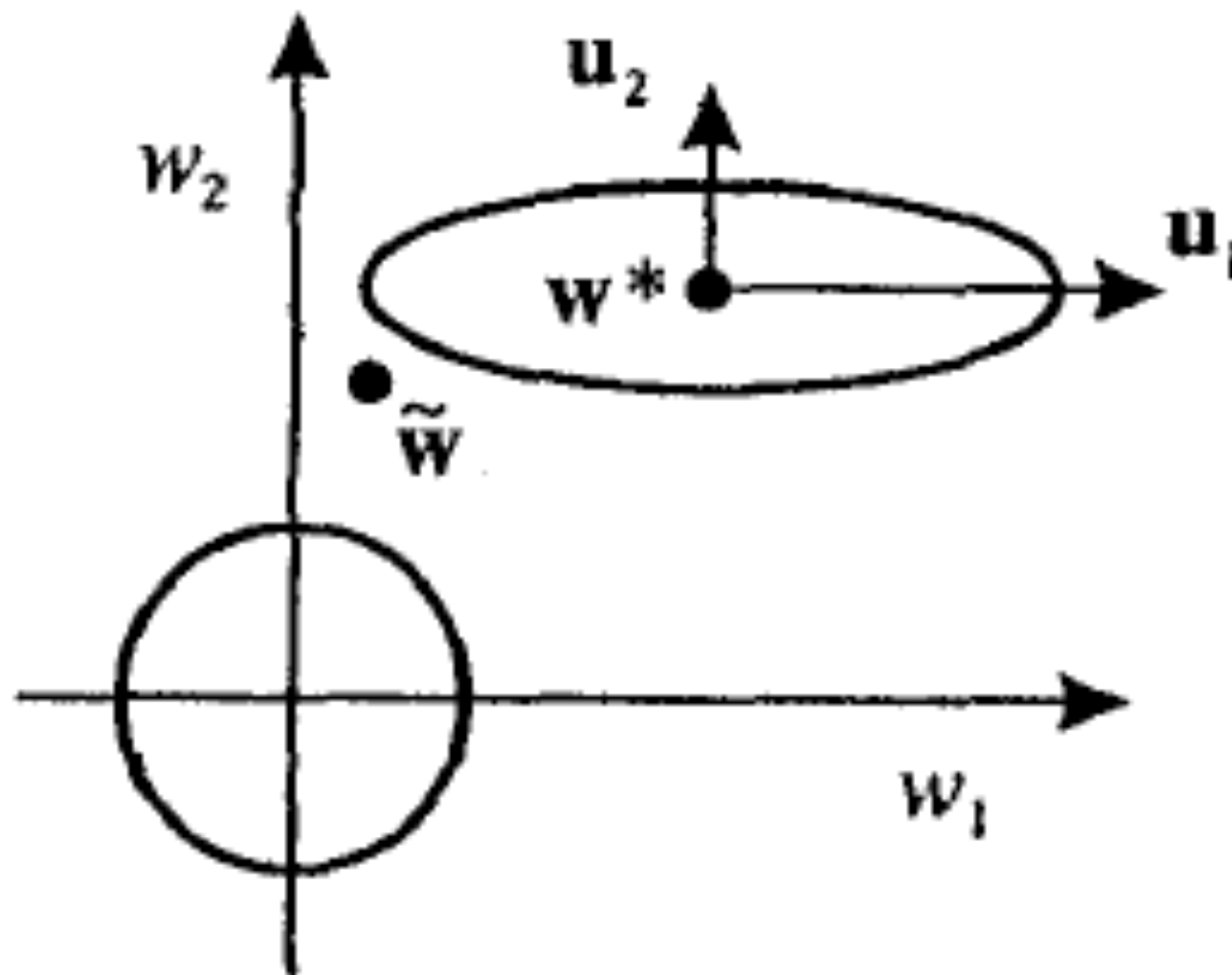
Underfit



Overfit

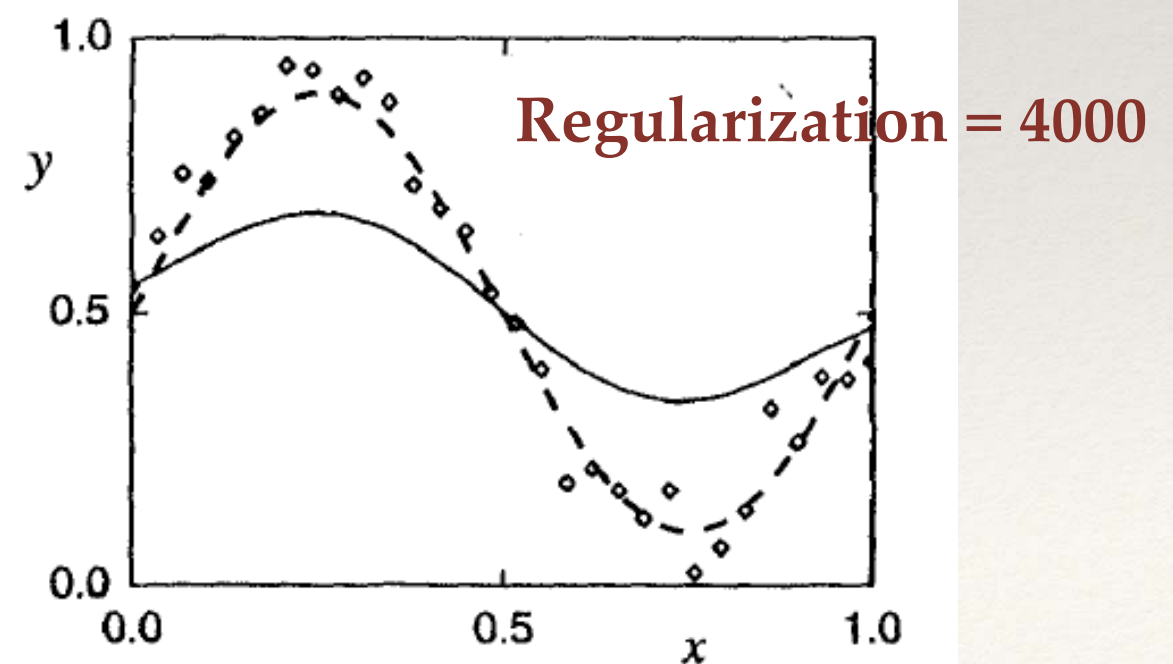
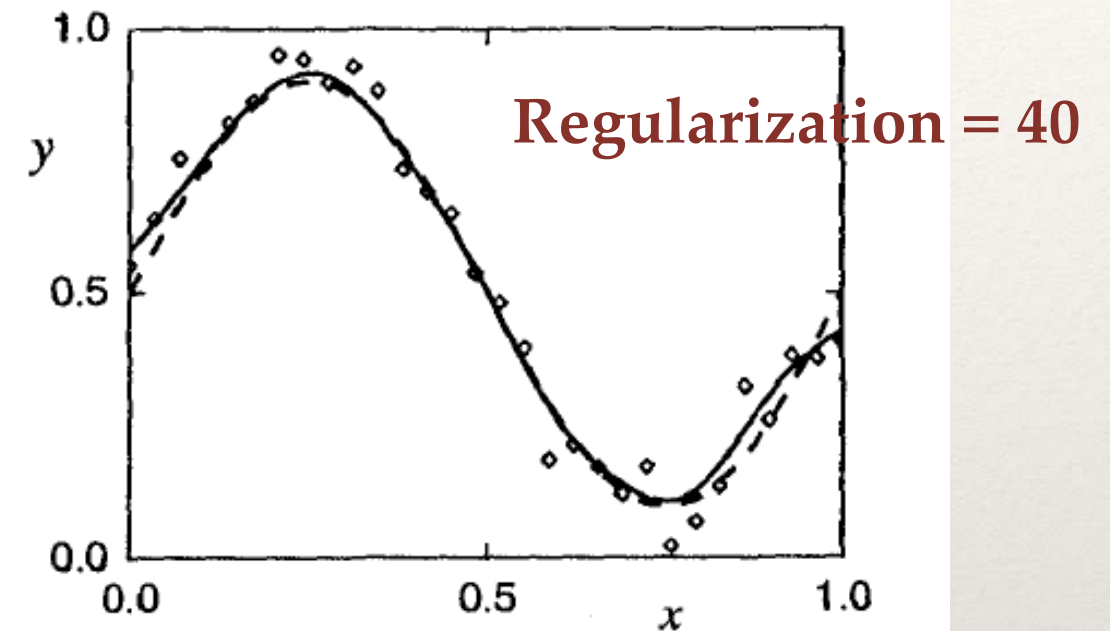
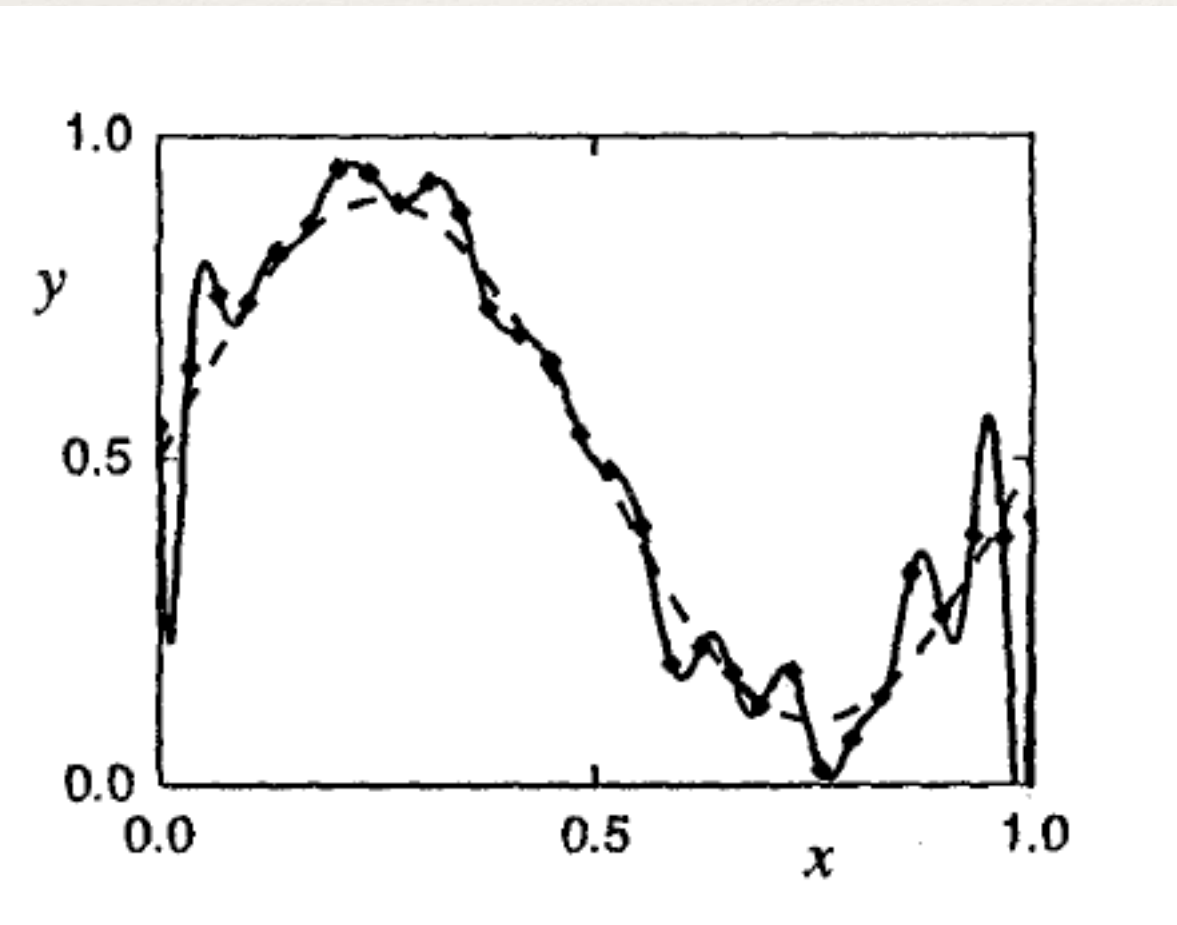


Weight Decay Based Regularization

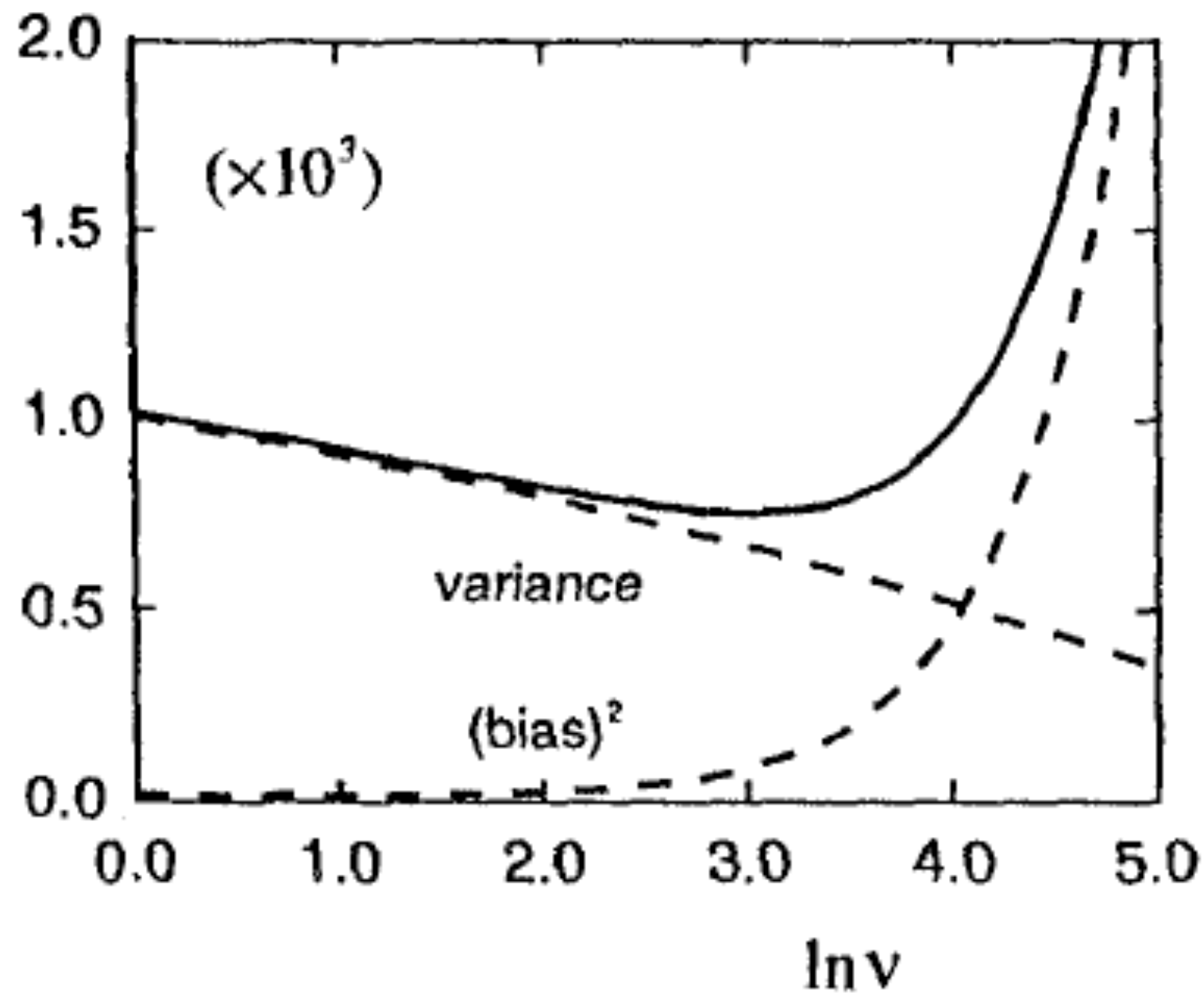


Weight Decay Regularization

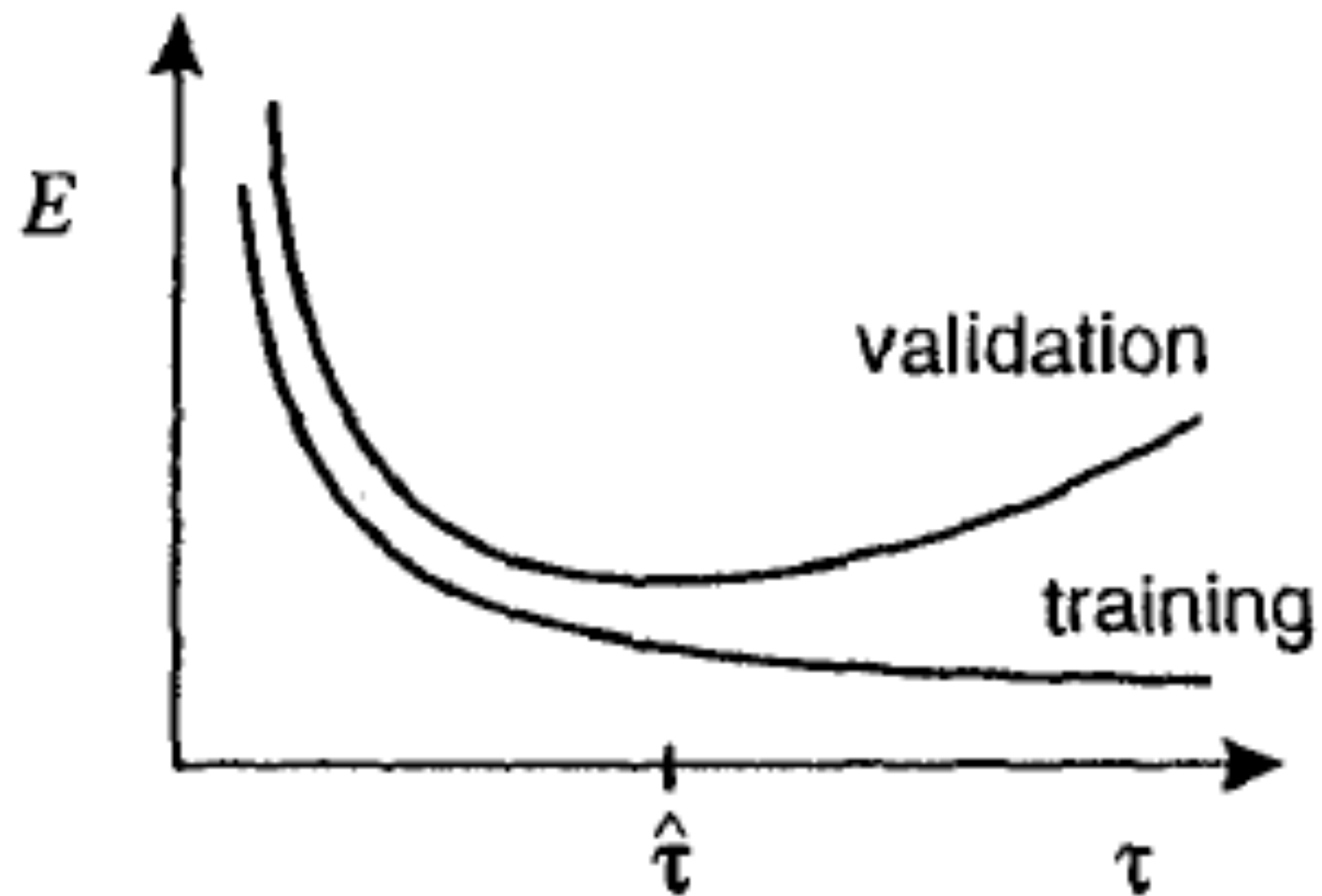
Regularization = 0



Regularization Effect on Learning



Early Stopping

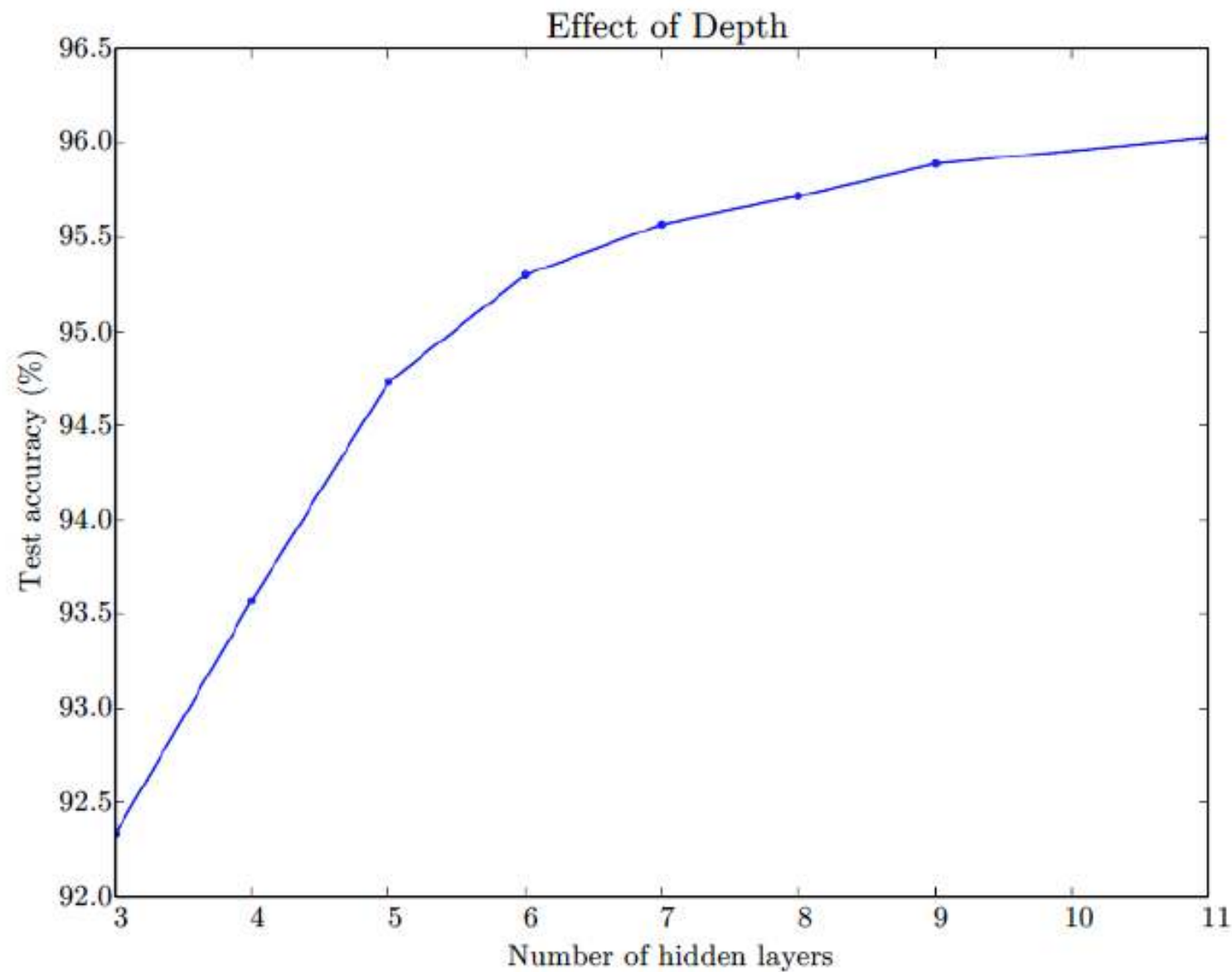


Most Popular in Practice

Neural Networks - Summary

- ❖ Details of Architecture
- ❖ Computation of gradient using back propagation.
- ❖ Error function and output layer activation
 - ❖ Neural networks estimate posterior probabilities
- ❖ Learning in Neural networks
 - ❖ Gradient descent - Properties
- ❖ Generalization of Neural Networks

Need for Depth



$$\mathbf{h}^{(1)} = g^{(1)} \left(\mathbf{W}^{(1)\top} \mathbf{x} + \mathbf{b}^{(1)} \right)$$

$$\mathbf{h}^{(2)} = g^{(2)} \left(\mathbf{W}^{(2)\top} \mathbf{h}^{(1)} + \mathbf{b}^{(2)} \right)$$

Need for Depth

